Systems of Linear Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Setting up the augmented matrix**

Write the augmented matrix for the system $2x + y - z = 3$, $x - 3y + 2z = -1$, $4x + y + z = 5$ (give it as its three rows).

Row 1: $\langle 2, 1, -1, 3 \rangle$. Row 2: $\langle 1, -3, 2, -1 \rangle$. Row 3: $\langle 4, 1, 1, 5 \rangle$ (each row lists the x, y, z coefficients, then the right-hand side).

2 **[Foundational] • Target: Setting up the augmented matrix**

Write the augmented matrix for the system $x - y = 4$, $3x + 2y = 7$.

Row 1: $\langle 1, -1, 4 \rangle$. Row 2: $\langle 3, 2, 7 \rangle$.

3 **[Foundational] • Target: Elementary row operations**

List the three types of elementary row operations.

1. Swap two rows. 2. Multiply a row by a nonzero scalar. 3. Add a multiple of one row to another row.

4 **[Foundational] • Target: Elementary row operations**

Given the augmented matrix with rows $\langle 1, 2, 5 \rangle$, $\langle 3, -1, 1 \rangle$, apply the row operation $R_2 \to R_2 - 3R_1$ and give the new Row 2.

$R_2 - 3R_1 = \langle 3, -1, 1 \rangle - 3\langle 1, 2, 5 \rangle = \langle 3 - 3, -1 - 6, 1 - 15 \rangle = \langle 0, -7, -14 \rangle$.

5 **[Foundational] • Target: Row echelon form and back-substitution**

A system's augmented matrix has been reduced to row echelon form with rows $\langle 1, 2, -1, 4 \rangle$, $\langle 0, 1, 3, -2 \rangle$, $\langle 0, 0, 1, 5 \rangle$ (variables x, y, z). Use back-substitution to solve.

From row 3: $z = 5$. From row 2: $y + 3(5) = -2 \Rightarrow y = -17$. From row 1: $x + 2(-17) - 5 = 4 \Rightarrow x - 39 = 4 \Rightarrow x = 43$. Solution: $(43, -17, 5)$.

6 **[Foundational] • Target: Row echelon form and back-substitution**

Explain why row echelon form makes a system easy to solve, in one or two sentences.

In row echelon form, the last nonzero row involves only the last variable, so it can be solved immediately; each preceding row then involves only variables already solved for plus one new one, letting the solution unwind one variable at a time — back-substitution.

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7 [Foundational] • Target: Inconsistent and dependent systems

Solve (or classify) the system $x + y = 3$, $2x + 2y = 8$.

The second equation is $2(x + y) = 8$, i.e., $x + y = 4$ — contradicting $x + y = 3$. The system is inconsistent (no solution): the two lines are parallel and distinct.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inconsistent and dependent systems

For the system $x - 2y = 4$, $-3x + 6y = k$, find the value of k that makes the system have infinitely many solutions (rather than no solution).

The second equation is -3 times the first row's coefficients ($-3(x - 2y) = -3x + 6y$), so it matches when $-3(4) = k$, i.e., $k = -12$. Any other k gives an inconsistent (no-solution) system.

9 [Intermediate] • Target: Why row operations preserve the solution set

Explain briefly why multiplying a row by a nonzero constant c doesn't change a system's solution set.

Multiplying both sides of an equation by a nonzero c produces an equivalent equation, since the operation is invertible — dividing by c recovers the original equation exactly.

10 [Intermediate] • Target: Why row operations preserve the solution set

Why is the restriction $c \neq 0$ needed for this row operation?

If $c = 0$, the row becomes the zero row, which is not invertible — the original row's information is destroyed, so the resulting system need not have the same solution set as the original.

11 [Intermediate] • Target: Why row operations preserve the solution set

Explain briefly why adding a multiple of one row to another row also preserves the solution set.

This operation is also invertible: subtracting the same multiple of the row back off recovers the original row exactly. Since any solution to the original system automatically satisfies the new (combined) equation too, and the operation can always be undone, the solution set is unchanged.

12 [Intermediate] • Target: Systems with a parameter

For what value(s) of k does the system $x + ky = 3$, $2x + 4y = 6$ have infinitely many solutions?

The second equation simplifies to $x + 2y = 3$. This matches the first equation exactly when $k = 2$ (giving two copies of the same line). For $k = 2$, infinitely many solutions; for any other value of k, the two lines are not parallel, so the system has a unique solution.

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13 [Intermediate] • Target: Systems with a parameter

For what value of k does the system $2x - y = 5$, $4x + ky = 1$ have NO unique solution (i.e., the coefficient matrix is singular)?

The system fails to have a unique solution exactly when the two equations' coefficients are proportional: $\frac{4}{2} = \frac{k}{-1} \Rightarrow 2 = -k \Rightarrow k = -2$.

14 [Intermediate] • Target: Application: mixture and rate problems

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 L of a 15% solution. Set up the system of equations (let x, y be the liters of each solution used).

Volume: $x + y = 20$. Acid content: $0.10x + 0.30y = 0.15(20) = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 1.01 Vectors in n Dimensions

Solve the mixture system set up above.

From the volume equation, $x = 20 - y$. Substituting: $0.10(20 - y) + 0.30y = 3 \Rightarrow 2 - 0.10y + 0.30y = 3 \Rightarrow 0.20y = 1 \Rightarrow y = 5$. Then $x = 15$. Use 15 L of the 10% solution and 5 L of the 30% solution.

16 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Two trains leave stations 300 km apart, heading toward each other. One travels at x km/h, the other at y km/h, and they meet after 2 hours: $2x + 2y = 300$. If the faster train's speed is $y = x + 20$, set up and solve the system for x and y.

Substituting $y = x + 20$ into $2x + 2y = 300$: $2x + 2(x + 20) = 300 \Rightarrow 4x + 40 = 300 \Rightarrow x = 65$. Then $y = 85$. Speeds: 65 km/h and 85 km/h.

17 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 4.01 Determinants

A theater sells adult tickets for 12 dollars and child tickets for 7 dollars. On a night with 200 tickets sold for a total of 1900 dollars, set up and solve the system for the number of adult (a) and child (c) tickets sold.

$a + c = 200$ and $12a + 7c = 1900$. From the first, $c = 200 - a$. Substituting: $12a + 7(200 - a) = 1900 \Rightarrow 5a + 1400 = 1900 \Rightarrow a = 100$. Then $c = 100$. So 100 adult tickets and 100 child tickets were sold.

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18 **[Challenging]** • **Target: Application: mixture and rate problems** • **Cross-Lesson: 1.01 Vectors in n Dimensions**

Explain why real-world word problems like the two above always translate into a LINEAR system (rather than, say, a quadratic one), given the quantities described.

Each condition (total count, total cost, total distance) is a simple sum or weighted sum of the unknowns — addition and constant multiplication only, with no products of two unknowns — so each equation is automatically linear in the unknowns.

19 **[Challenging]** • **Target: Setting up the augmented matrix** • **Cross-Lesson: 3.01 Matrices & Matrix Algebra**

Level 1 · FoundationsConvert the matrix into Reduced Row Echelon Form (RREF): $M = (1, -2, 1; 2, -4, 5)$.

Step 1 · Eliminate below pivot 1: Perform $R_2 \leftarrow R_2 - 2R_1$: $(1, -2, 1; 0, 0, 3)$. Step 2 · Scale pivot 2 to 1: Perform $R_2 \leftarrow \frac{1}{3}R_2$: $(1, -2, 1; 0, 0, 1)$. Step 3 · Eliminate above pivot 2: Perform $R_1 \leftarrow R_1 - R_2$: $(\mathbf{1, -2, 0; 0, 0, 1})$.

20 **[Challenging]** • **Target: Elementary row operations** • **Cross-Lesson: 4.01 Determinants**

Level 3 · Advanced / University RigorFind the general solution in parametric vector form for the underdetermined linear system: $x_1 + 2x_2 - 3x_3 + x_4 + 2x_5, = 3; -x_1 - 2x_2 + 4x_3 - 2x_4 - x_5, = -2; 2x_1 + 4x_2 - 5x_3 + x_4 + 5x_5, = 7$

Step 1 · Set up Augmented Matrix and Eliminate: x Perform $R_3 \leftarrow R_3 - R_2$ and $R_1 \leftarrow R_1 + 3R_2$ to reach RREF: $(1, 2, 0, -2, 5, |, 6; 0, 0, 1, -1, 1, |, 1; 0, 0, 0, 0, 0, |, 0)$. Step 2 · Identify Basic and Free Variables: Pivots in columns 1 and 3 $\Rightarrow x_1, x_3$ are basic. Columns 2, 4, 5 lack pivots $\Rightarrow x_2 = r, x_4 = s, x_5 = t$ are free parameters. Step 3 · Express Basic Variables: $x_1 = 6 - 2r + 2s - 5t, \qquad x_3 = 1 + s - t.$ Step 4 · Parametric Vector.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Setting up the augmented matrix

Consider the system of linear equations represented by the following augmented matrix: $(1, 2, |, 5; 0, 1, |, 2)$ Use row reduction to determine the unique solution for x and y in the system. What is the value of x?

a. 1

b. 2

c. **3** ✓

d. 4

The augmented matrix represents the system of equations: 1) $x + 2y = 5$ 2) $y = 2$ Since the second equation directly gives the value of y as 2, we can substitute this into the first equation to find x. Substitute $y = 2$ into equation (1): $x + 2(2) = 5$ $x + 4 = 5$ $x = 5 - 4$ $x = 1$ Therefore, the unique solution is $x = 1$ and $y = 2$. The question asks for the value of x.

22 [Foundational] • Target: Elementary row operations

Consider the following system of linear equations represented by an augmented matrix: $[1, 2, |, 5; 0, 1, |, 2]$. What is the value of x in the first equation?

a. 1

b. 2

c. **3** ✓

d. 4

The augmented matrix represents the system of equations: $x + 2y = 5$ and $y = 2$. We are asked to find the value of x. We can use substitution to solve for x. Substitute the value of y from the second equation into the first equation: $x + 2(2) = 5$. This simplifies to $x + 4 = 5$. Subtracting 4 from both sides gives $x = 1$. Therefore, the value of x in the first equation is 1.

23 [Foundational] • Target: Solving 2x2 systems by elimination

Solve for z: $x + y + z = -1$, $y + z = -4$, $z = 1$ (already row-echelon).

a. **1** ✓

b. 3

c. -5

d. 2

Back-substitute: $z = 1$; $y = -4 - (1) = -5$; $x = -1 - (-5) - (1) = 3$.

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24 [Foundational] • Target: Solving 3x3 systems by Gaussian elimination

Consider the matrix $A = (1, 1, 0; 1, 0, 1; 0, 1, 1)$. Let $\mathbf{b} = (1; 1; 1)$. We want to find the vector $\hat{\mathbf{x}}$ that minimizes $Ax - b$. This is equivalent to finding the projection of $\mathbf{b}$ onto the column space of A, denoted by $\text{Col}(A)$. The normal equations for this least squares problem are $A^T A\mathbf{x} = A^T\mathbf{b}$. The unique solution $\hat{\mathbf{x}}$ to the normal equations yields the best approximate solution. Calculate the value of $\hat{\mathbf{x}}_1$, the first component of $\hat{\mathbf{x}}$.

a. 1/3

b. **1/2 ✓**

c. 2/3

d. 3/4

The normal equations are given by $A^T A\mathbf{x} = A^T\mathbf{b}$. First, we compute $A^T A$ and $A^T\mathbf{b}$. \\
$A^T = (1, 1, 0; 1, 0, 1; 0, 1, 1)$ \\ $A^T A = (1, 1, 0; 1, 0, 1; 0, 1, 1)(1, 1, 0; 1, 0, 1; 0, 1, 1) = (2, 1, 1; 1, 2, 1; 1, 1, 2)$ \\
$A^T\mathbf{b} = (1, 1, 0; 1, 0, 1; 0, 1, 1)(1; 1; 1) = (2; 2; 2)$ \\ So, we need to solve the system.

25 [Foundational] • Target: Row echelon form and back-substitution

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (2, k\ 1, 3)$ with respect to the standard basis. The image of the vector $\mathbf{v} = (1\ 1)$ under T is $T(\mathbf{v}) = (3\ 4)$. Find the value of the parameter k.

a. **1 ✓**

b. 2

c. 3

d. 4

The linear transformation T is represented by the matrix $A = (2, k\ 1, 3)$. The action of T on a vector $\mathbf{v}$ is given by $T(\mathbf{v}) = A\mathbf{v}$. We are given that $\mathbf{v} = (1\ 1)$ and $T(\mathbf{v}) = (3\ 4)$. Therefore, we have the matrix equation: $(2, k\ 1, 3)(1\ 1) = (3\ 4)$ Performing the matrix multiplication on the left side gives: $(2(1) + k(1)\ 1(1) + 3(1)) = (2 + k\ 4)$ Equating this to.

26 [Foundational] • Target: Inconsistent and dependent systems

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (1, 2; k, 4)$. The image of the vector $\mathbf{v} = (1; 1)$ under this transformation is $T(\mathbf{v}) = (3; 5)$. Determine the value of the parameter k.

a. **1 ✓**

b. 2

c. 3

d. 4

The linear transformation T is defined by $T(\mathbf{x}) = A\mathbf{x}$. We are given $A = (1, 2; k, 4)$ and $\mathbf{v} = (1; 1)$. The image of $\mathbf{v}$ is $T(\mathbf{v}) = A\mathbf{v} = (1, 2; k, 4)(1; 1) = (1 \cdot 1 + 2 \cdot 1; k \cdot 1 + 4 \cdot 1) = (3; k + 4)$. We are given that $T(\mathbf{v}) = (3; 5)$. Equating the components, we have $3 = 3$ (which is consistent) and $k + 4 = 5$. Solving for k, we get.

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27 [Foundational] • Target: Why row operations preserve the solution set

How many solutions does the system $3x + (3)y = 6, -6x + (-6)y = -13$ have?

a. no solution ✓

b. exactly one solution

c. infinitely many solutions

d. exactly two solutions

The coefficients are proportional but the constants aren't — parallel, distinct lines, so no point satisfies both (inconsistent).

28 [Foundational] • Target: Systems with a parameter

Consider a matrix $A = (2, -1; 3, k)$. If the determinant of the matrix A^2 is 100, find the positive value of the parameter k.

a. 4 ✓

b. 5

c. 2

d. 8

The determinant of a matrix $A = (a, b; c, d)$ is given by $\det(A) = ad - bc$. For the given matrix $A = (2, -1; 3, k)$, the determinant is $\det(A) = (2)(k) - (-1)(3) = 2k + 3$. A fundamental property of determinants states that $\det(A^n) = (\det(A))^n$. Therefore, for A^2, we have $\det(A^2) = (\det(A))^2$. We are given that $\det(A^2) = 121$. Substituting the determinant of A, we get $(2k + 3)^2 = 121$. Taking the square root of both sides yields $2k + 3 = \pm\sqrt{121}$, which means.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Application: mixture and rate problems

Consider a linear transformation represented by the matrix $A = (2, 1; 1, -2)$. A point P is mapped to P' by this transformation. If the coordinates of P are $(x; y)$ and the coordinates of P' are $(x'; y')$, then $(x'; y') = A(x; y)$. Suppose a curve is defined by the equation $x'^2 + y'^2 = 5$. What is the equation of the original curve $x^2 + y^2 = k$ in terms of x and y that transforms into this circle? Find the value of k.

a. 1

b. **2 ✓**

c. 3

d. 4

The transformation is given by $(x'; y') = (2, 1; 1, -2)(x; y)$. We can express $(x; y)$ in terms of $(x'; y')$ by finding the inverse of matrix A. The determinant of A is $\det(A) = (2)(-2) - (1)(1) = -4 - 1 = -5$. The inverse matrix is $A^{-1} = \frac{1}{-5}(-2, -1; -1, 2) = (2/5, 1/5; 1/5, -2/5)$. Therefore, $(x; y) = A^{-1}(x'; y') = (2/5, 1/5; 1/5, -2/5)(x'; y')$. This giv.

30 [Intermediate] • Target: Setting up the augmented matrix

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x, y) = (3x - y, x + 2y)$. Consider the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for $\mathbb{R}^2$, where $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$. Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for $\mathbb{R}^2$, where $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -1)$. If the matrix representation of T with respect to the basis $\mathcal{B}$ is $[T]_{\mathcal{B}} = (a, b\ c, d)$, and $a + d = 5$, determine the value of $\det(T)$.

a. **7 ✓**

b. 5

c. 3

d. 2

The matrix representation of T with respect to the standard basis $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$ is $A = (3, -1\ 1, 2)$. The determinant of T is the determinant of its matrix representation with respect to any basis, so $\det(T) = \det(A) = (3)(2) - (-1)(1) = 6 + 1 = 7$. The information about the basis $\mathcal{B}$ and $[T]_{\mathcal{B}} = (a, b\ c, d)$ with $a + d = 5$ is a distractor. The trace of a matrix is invariant under a change of basis, so.

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31 **[Intermediate]** • **Target: Elementary row operations**

A solid object in $\mathbb{R}^3$ is defined by the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, and the plane passing through the points $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$. If a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ is represented by the matrix $M = (1, 0, 0; 0, 1, 0; 0, 0, 2)$, what is the volume of the transformed solid object?

a. 12

b. **24** ✓

c. 36

d. 48

The volume of the solid region bounded by the coordinate planes and the plane passing through $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$ can be calculated. The equation of the plane passing through these points is given by $\frac{x}{6} + \frac{y}{4} + \frac{z}{3} = 1$. This is a tetrahedron with vertices at $(0, 0, 0)$, $(6, 0, 0)$, $(0, 4, 0)$, and $(0, 0, 3)$. The volume of such a tetrahedron is $V_{original} = \frac{1}{6}|abc|$, where a, b, c are the intercepts on the axes. Thus.

32 **[Intermediate]** • **Target: Solving 2x2 systems by elimination**

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (2, 0\ 0, 3)$. What is the image of the vector $\mathbf{v} = (1\ 1)$ under this transformation, i.e., find $T(\mathbf{v})$?

a. (1 1)

b. (2 3) ✓

c. (3 2)

d. (4 9)

The linear transformation T is represented by the matrix A. Therefore, the image of a vector $\mathbf{v}$ under T is given by $T(\mathbf{v}) = A\mathbf{v}$. In this case, $A = (2, 0\ 0, 3)$ and $\mathbf{v} = (1\ 1)$. So, $T(\mathbf{v}) = (2, 0\ 0, 3)(1\ 1) = (2 \times 1 + 0 \times 1; 0 \times 1 + 3 \times 1) = (2\ 3)$.

33 **[Intermediate]** • **Target: Solving 3x3 systems by Gaussian elimination**

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (1, 2; 3, 4)$. If a vector $\mathbf{v} = (2; 1)$ is transformed by T, what is the resulting vector $T(\mathbf{v})$?

a. (4; 10) ✓

b. (2; 3)

c. (4; 7)

d. (3; 5)

The linear transformation T can be applied to a vector $\mathbf{v}$ by multiplying the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = (1, 2; 3, 4)(2; 1)$ To perform this matrix-vector multiplication, we take the dot product of each row of the matrix with the vector: Row 1: $(1)(2) + (2)(1) = 2 + 2 = 4$ Row 2: $(3)(2) + (4)(1) = 6 + 4 = 10$ Therefore, the resulting vector is $(4; 10)$.

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34 [Intermediate] • Target: Row echelon form and back-substitution

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = (2, 1; 1, 3)$. If $\mathbf{v} = (1; 2)$, find $T(\mathbf{v})$.

a. $(4; 7)$ ✓

b. $(3; 7)$

c. $(4; 5)$

d. $(5; 7)$

To find $T(\mathbf{v})$, we multiply the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = (2, 1; 1, 3)(1; 2)$. Performing the matrix-vector multiplication: $T(\mathbf{v}) = ((2 \times 1) + (1 \times 2); (1 \times 1) + (3 \times 2)) = (2 + 2; 1 + 6) = (4; 7)$. Thus, the correct answer is $(4; 7)$.

35 [Intermediate] • Target: Inconsistent and dependent systems

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, 3 \rangle$.

a. $[(1, 3; 3, 9)]$

b. $\left[\left(\frac{9}{10}, -\frac{3}{10}; -\frac{3}{10}, \frac{1}{10}\right)\right]$

c. $\left[\left(-\frac{4}{5}, \frac{3}{5}; \frac{3}{5}, \frac{4}{5}\right)\right]$

d. $\left[\left(\frac{1}{10}, \frac{3}{10}; \frac{3}{10}, \frac{9}{10}\right)\right]$ ✓

$P = \frac{\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}} = \frac{1}{10}[(1, 3; 3, 9)] = \left[\left(\frac{1}{10}, \frac{3}{10}; \frac{3}{10}, \frac{9}{10}\right)\right]$.

36 [Intermediate] • Target: Why row operations preserve the solution set

Which matrix represents the rotation counterclockwise by 60°?

a. $\left[\left(\frac{1}{2}, \frac{\sqrt{3}}{2}; -\frac{\sqrt{3}}{2}, \frac{1}{2}\right)\right]$

b. $\left[\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}; \frac{\sqrt{3}}{2}, \frac{1}{2}\right)\right]$ ✓

c. $\left[\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}; -\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)\right]$

d. $[(0, -1; 1, 0)]$

Find the images of $\mathbf{e}_1, \mathbf{e}_2$: $\langle \frac{1}{2}, \frac{\sqrt{3}}{2} \rangle$ and $\langle -\frac{\sqrt{3}}{2}, \frac{1}{2} \rangle$. These are the columns.

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37 [Intermediate] • Target: Systems with a parameter

Which matrix represents the reflection across the x-axis?

a. $[(-1, 0; 0, 1)]$

b. $[(0, -1; 1, 0)]$

c. $[(1, 0; 0, -1)]$ ✓

d. $[(1, 0; 0, 1)]$

Find the images of $\mathbf{e}_1, \mathbf{e}_2$: $\langle 1, 0 \rangle$ and $\langle 0, -1 \rangle$. These are the columns.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 1.01 Vectors in n Dimensions

Which matrix represents the reflection across the y-axis?

a. $[(-1, 0; 0, 1)]$ ✓

b. $[(1, 0; 0, -1)]$

c. $[(0, -1; 1, 0)]$

d. $[(1, 0; 0, 1)]$

Find the images of $\mathbf{e}_1, \mathbf{e}_2$: $\langle -1, 0 \rangle$ and $\langle 0, 1 \rangle$. These are the columns.

39 [Challenging] • Target: Setting up the augmented matrix • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Which matrix represents the reflection across the line y = x?

a. $[(0, -1; -1, 0)]$

b. $[(0, 1; 1, 0)]$ ✓

c. $[(0, -1; 1, 0)]$

d. $[(1, 0; 0, 1)]$

Find the images of $\mathbf{e}_1, \mathbf{e}_2$: $\langle 0, 1 \rangle$ and $\langle 1, 0 \rangle$. These are the columns.

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40 [Challenging] • Target: Elementary row operations • Cross-Lesson: 4.01 Determinants

Which matrix represents the orthogonal projection onto the x-axis?

a. $[(1, 0; 0, 0)]$ ✓

b. $[(-1, 0; 0, 0)]$

c. $[(0, 0; 0, 1)]$

d. $[(0, -1; 1, 0)]$

Find the images of $\mathbf{e}_1, \mathbf{e}_2$: $\langle 1, 0\rangle$ and $\langle 0, 0\rangle$. These are the columns.

41 [Challenging] • Target: Solving 2x2 systems by elimination • Cross-Lesson: 1.01 Vectors in n Dimensions

Which matrix represents the horizontal shear x ■ x + 3y?

a. $[(1, 0; 3, 1)]$

b. $[(-1, -3; 0, -1)]$

c. $[(1, 3; 0, 1)]$ ✓

d. $[(0, -1; 1, 0)]$

Find the images of $\mathbf{e}_1, \mathbf{e}_2$: $\langle 1, 0\rangle$ and $\langle 3, 1\rangle$. These are the columns.

42 [Challenging] • Target: Solving 3x3 systems by Gaussian elimination • Cross-Lesson: 3.01 Matrices & Matrix Algebra

S has matrix $[(2, -3; 0, -3)]$ and T has matrix $[(3, -1; -3, -2)]$. Find the matrix of $S \circ T$ (apply T first).

a. $[(15, 4; 9, 6)]$ ✓

b. $[(6, -6; -6, 15)]$

c. $[(5, -4; -3, -5)]$

d. $[(6, 3; 0, 6)]$

$(S \circ T)(\mathbf{x}) = A(B\mathbf{x})$, so the matrix is $AB = [(15, 4; 9, 6)]$.

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43 [Challenging] • Target: Row echelon form and back-substitution • Cross-Lesson: 4.01 Determinants

Find the standard matrix of $T(x, y) = \langle -2x + 3y, -3x \rangle$.

a. $[(-2, -3; 3, 0)]$

b. $[(-2, 3; -3, 0)]$ ✓

c. $[(2, -3; 3, 0)]$

d. $[(-3, 0; -2, 3)]$

The columns are $T(\mathbf{e}_1) = \langle -2, -3 \rangle$ and $T(\mathbf{e}_2) = \langle 3, 0 \rangle$.

44 [Challenging] • Target: Inconsistent and dependent systems • Cross-Lesson: 1.01 Vectors in n Dimensions

For which k are $\langle -2, -4 \rangle$ and $\langle 1, k \rangle$ linearly dependent?

a. 3

b. -2

c. 4

d. 2 ✓

They are dependent exactly when $\det = 4 - 2k = 0$, i.e. $k = 2$.

45 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Let A be a 2×2 matrix given by $A = (2, k; 3, -1)$. If $A^2 = (1, -7; 3, 4)$, find the value of k.

a. 2

b. **3** ✓

c. 4

d. 5

We are given the matrix $A = (1, 2; k, 3)$. The transpose of A, denoted by A^T, is obtained by interchanging its rows and columns. Thus, $A^T = (1, k; 2, 3)$ We are given the equation $A + A^T = (2, 5; 5, 6)$. Let's compute the sum $A + A^T$: $A + A^T = (1, 2; k, 3) + (1, k; 2, 3) = (1 + 1, 2 + k; k + 2, 3 + 3) = (2, 2 + k; k + 2, 6)$ Now, we equate this resulting matrix to the given matrix $A + A^T = (2, 5; 5, 6)$.

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